E. Mahima [192212227]

ITA0329 - MOBILE COMPUTING

CIA-1

4) Sol:

Given,

→ Transfers medical imaging files = 240

→ size = 18 MB

→ Data rate = 20 Kbps

→ Reduces capacity by → 18%

→ Effective throughput = 85%

→ Retransmission rate = 12%

→ UMTS transfer time = 384 Kbps

Allocated timeslots = 4

→ mobility related interruptions = 8

→ lasting → 2.5 sec

(i) Allocated time slots x Data rate = reduction in transfer time

$R = 4 \times 20 Kbps$

$R = 80 Kbps$

(ii) GPRS emergency deadline;

$\Rightarrow T \times (1 - capacity) \times Effective\ throughput$

$\Rightarrow 80 \times (1 - 0.18) \times 0.85 \times 10^3$

$\Rightarrow 80 \times (0.82) \times 0.85 \times 10^3$

$\Rightarrow 55.76 Kbps$

→ transfer medical files x size

$\Rightarrow$

$\Rightarrow$ 240 x 18 MB

$\Rightarrow$ 4320 MB

→ GPRS interruptions,

$\Rightarrow$ 4320 x mobility related interruptions

$\Rightarrow$ $4320 \times 8 \times 10^6$

$\Rightarrow$ 34560×10^6

$\Rightarrow$ 34560 Mbps

→ GPRS interruptions x packet retransmission

$= 34560 \times 1.12$

$\Rightarrow$ 38707.2.

→ Transfer time achieved by UMTS,

$$T = \frac{\text{Data rate}}{\text{Throughput}}$$

$$T = \frac{20\text{kbps}}{0.85}$$

$$T = \frac{20 \times 10^3}{0.85}$$

$$\therefore \boxed{T = 23.529\text{s}}$$

GPRS meets 12-hour deadline,

$\frac{12}{38707.2}$ x 4 timeslots $\Rightarrow$ 1.240ms //

1)

Sol:

Given,

Area = 1,200km²

cell radius = 1.8km

Available spectrum = 48MHz & 200kHz

cluster sizes N=7 & N=4

Reduction = 15%

To calculate:

(i) No.of cells = $\frac{1200 km^2}{200 * Hz}$

∴ No.of cells = 6

(ii) Total available channels,

N=7 & N=4 ⇒ $\frac{6 * 7}{1.8km}$ ⇒ Cell radius x available spectrum

⇒ 1.8 km x 200kHz

⇒ 360M channels

(iii) Theoretical effective channels per cell,

⇒ available channels x No.of cells

⇒ 2160 x 10^6 channels

⇒ 2160M channels

(iv) co-channel reuse distances,

N=7, $\frac{7 \times 1200 km^2}{0.15}$ ⇒ 56000km

N=4, $\frac{4 \times 1200 km^2}{0.15}$ ⇒ 32000km

2) Sol:

Given,

Time = 40minutes

cell boundaries = 15

Distance = 90km/h

intra-PLMN Handoff requires ⇒ 120ms for BSS coordination

⇒ 80ms for NSS path switching

while,

each inter-PLMN requires,

⇒ 350 ms for authentication

⇒ 250ms for roaming registration

⇒ 400ms for recovery

Data loss = 512 kbps

(i) convert min → sec,

40min → sec = 40×60

= 2400sec

(ii) Normal Hand-off Delay,

⇒ (120 + 80)ms − $\frac{90km/h}{2400}$

185ms − $\frac{2400}{90000}$ 37.5

⇒ 37.315 delay

(iii) Additional delay,

350ms + $\frac{90km/h}{2400sec}$

⇒ 37.5 − 0.35

⇒ 37.15 → Additional delay

iv) Roaming delay,

(250 + 350)ms + $\frac{Distance}{Time}$.

600ms + $\frac{90km/h}{2400}$

600ms + 37.5

→ 637.5 roaming delay

v) Recovery delay,

$\frac{90km/h}{2400}$ − 400ms

37.5 − 0.40ms

⇒ 37.1 delay

3. (i) Given,

Base transceiver Stations = 12 — 3 → sectors, 4 → Transceivers

TDMA time slots = 8

Timeslots per sector = 2

each sector serves = 1200 subscribers

call holding time = 90 sec

Abis Interface = 16Kbps

Single E1 link = 2.048 Mbps

Channel utilization limit = 80%

To determine,

(i) Minimum no of TRXs required per sector,

$$C = (TRX_S \times 8) - 2$$

$$C = (4 \times 8) - 2$$

$$C = 32 - 2 \Rightarrow \boxed{C = 30}$$

(ii) To verify,

Whether a single E1 Abis link remains sufficient after the upgrade,

$$\Rightarrow \frac{2.048\ Mbps}{16kbps} - (80\%)$$

$$\Rightarrow \frac{2.048 \times 10^6}{16 \times 10^3} - (0.80)$$

$$\Rightarrow \frac{2.048 \times 10^3}{16} - (0.80) \Rightarrow \frac{2048}{16} - 0.80$$

$$\Rightarrow 128 - 0.80 \Rightarrow \underline{127.2}$$

5) (E) Given,

Distributed software = 120GB

Mobile devices = 2000

GSM provides = 9.6 Kbps

effective throughput = 64Kbps

retransmission = 10%

5G provides = 25Mbps

protocol overhead = 4%

devices transfer data = 40%

1GB = 1000MB

1MB = 8 megabits

(i) Data volume per device. → Individual transfer time,

$$\Rightarrow \frac{120\,GB}{2000}$$

Converting GB to MB,

$$\frac{120 \times 1000}{2000} \Rightarrow 60\,MB$$

Converting to megabits

$$\Rightarrow 60 \times 8 = 480$$

→ 176.9 Mbps × (0.10 + 0.04)

⇒ 176.9 Mbps × 0.14

⇒ 24.766 Mbps

⇒ 480 × 9.6 kbps

⇒ 4608 kbps

→ 4608 kbps × (64kbps)

(1 − 0.40) × 64 kbps

⇒ 4608 × 10^3 (0.60) × 64 × 10^3

⇒ 176947.2 × 10^6 bps

⇒ 176.9 Mbps

⇒ $T = \frac{0.40}{64\,kbps}$

∴ $T = 6.25\,\mu s$